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===== PART 1: Load geometry & identify faces

[illegible]

===== PART 2 =====

```
fixedFace = 1;      % from face plot
loadFace  = 3;
loadDir   = [0; -1; 0];

E = 200e3;      % adjust units to match geometry scale (see note below)
nu = 0.30;
Sy = 250;

structuralProperties(model, "YoungsModulus", E, "PoissonsRatio", nu);
structuralBC(model, "Face", fixedFace, "Constraint", "fixed");

A = 15*2
P = 120/A;
structuralBoundaryLoad(model, "Face", loadFace, "SurfaceTraction", P*loadDir);
```

```

% ---- Mesh sized relative to actual geometry, not fixed numbers ----
diagLen = norm(dims);           % diagonal of bounding box, real units
Hmax = diagLen/25;              % ~25 elements across the part
Hmin = diagLen/150;            % finer near small features like the bore

generateMesh(model,"Hmax",Hmax,"Hmin",Hmin);

R = solve(model);

figure;
pdeplot3D(model,"ColorMapData",R.VonMisesStress,"Deformation",R.Displacement)
;
title("Von-Mises Stress"); colorbar;

maxStress = max(R.VonMisesStress);
fprintf("Max Von-Mises Stress = %.4f\n", maxStress);
fprintf("Safety Factor = %.2f\n", Sy/maxStress);

```

===== Results =====

```

figure;
pdeplot3D(model,"ColorMapData",R.VonMisesStress,"Deformation",R.Displacement)
;
title("Von-Mises Stress (MPa)"); colorbar;

maxStress = max(R.VonMisesStress);
maxDisp = max(sqrt(sum(R.Displacement.ux.^2 + R.Displacement.uy.^2 +
R.Displacement.uz.^2,1)));

fprintf("Max Von-Mises Stress = %.2f MPa\n", maxStress);
fprintf("Max Displacement      = %.4f mm\n", maxDisp);
fprintf("Safety Factor          = %.2f\n", Sy/maxStress);

```

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